

## Human Review: Sharp Constant for Proposition P:schur

Original automatic proof: `solution_packet.pdf`

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Status: Verified.

Verdict: The proof appears correct. It gives a sharpness example showing that, for  $c \geq 1$ , the constant  $2c + 1$  in Kalenda's implication

$$(iii_c) \implies (ii_{2c+1})$$

cannot be improved.

## Human Comments

The construction is well targeted. For  $c \geq 1$ , the proof considers

$$X_c = \mathbb{K} \oplus \ell_1$$

with norm

$$\|(t, x)\|_c = \max\{\|x\|_1, |t| + 2c\|x\|_\infty\}.$$

The sequence

$$y_n = (1, e_n)$$

then has

$$\|y_n\|_c = 2c + 1.$$

On the other hand, the averaging argument shows that every weak-star cluster point of  $(y_n)$  in  $X_c^{**}$  has norm exactly 1. Hence condition  $(i_C)$  fails for every  $C < 2c + 1$ . Since Proposition P:schur in the source paper proves  $(i_C) \iff (ii_C)$ , this also gives failure of  $(ii_C)$  for every  $C < 2c + 1$ . This proves sharpness in the formulation of Question 9.2.

The proof that  $X_c$  satisfies  $(iii_c)$  also looks correct. The key point is to take an arbitrary bounded sequence  $z_n = (t_n, x_n)$ , pass to a subsequence with  $t_n \rightarrow t$ ,  $x_n$  converging coordinatewise to some  $x \in \ell_1$ , and  $\|x_n - x\|_1 \rightarrow A$ . The fact that the coordinatewise limit belongs to  $\ell_1$  follows from Fatou's lemma, after the usual diagonal extraction. The gliding-hump lemma then gives

$$\delta_{\ell_1}(x_n - x) \geq 2A$$

and, after passing to a further subsequence, the estimates

$$\|u_p - u_q\|_1 \leq 2A + \varepsilon, \quad \|u_p - u_q\|_\infty \leq A + \varepsilon,$$

where  $u_n = x_n - x$ . These estimates give

$$\tilde{\text{ca}}(z_n) \leq 2cA \leq c\delta_{X_c}(z_n),$$

which is precisely  $(iii_c)$ .

The only improvement I would suggest is in the proof of Lemma 1. The argument uses the fact that  $\ell_1$  has the 1-Schur property, namely

$$\text{ca}(w_n) \leq \delta_{\ell_1}(w_n)$$

for bounded sequences in  $\ell_1$ . This is a standard quantitative Schur fact, but it should ideally be cited explicitly. In the source paper, this is recalled in Example 5.9, where  $\ell_1$  is said to have the 1-Schur property by Kalenda–Spurný, Theorem 1.3. Adding this citation at the use in Lemma 1 would make the proof more self-contained and easier to check.

Aside from this minor citation/reference issue, the main construction, the verification of  $(iii_c)$ , the cluster-point norm calculation, and the final sharpness argument all look sound.

## Review Outcome

Accepted as an essentially correct solution to the optimality question for  $c \geq 1$ . The small-constant range  $0 < c < 1$  is also plausibly handled by the Rosenthal–James argument in the packet, but the main contribution for Question 9.2 is the sharp  $2c + 1$  example above.